

PAPER_1738 — Quantum Complexity BQP/P Bound = $BQP/P \leq 2^{(D_{\text{phys}}/2)} = 2^2 = 4$ per oracle level

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Quantum Info **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_1271-1299 broader corpus)

Observation

PAPER_1298 (PAPER_1271-1299 broader corpus) gives:

$BQP/P \leq 2^{(D_{\text{phys}}/2)} = 2^2 = 4$ per oracle level

Status: **EXACT**.

UQFF Closed Identity

$BQP/P \leq 2^{(D_{\text{phys}}/2)} = 2^2 = 4$ per oracle level EXACT

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1298
- Related: PAPER_1716-1725 (tier-30 Millennium); PAPER_1726-1735 (tier-31 neutron lifetime)
- Calculator dispatch: `calculate_paradox({"paradox": "bqp_bound_2_pow_d_2"})`

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